WELCOME

MongoDB SI Certification Presentation

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Agenda

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Customer Overview

Universidad Peruana de Ciencias Aplicadas (UPC) is an educational institution. it is the first global university in Peru. At UPC, there are more than 82.228 enrolled students. In 2016, it became the first and only Peruvian university to obtain institutional accreditation by WASC Senior College and University Commission.

Since its creation in 1994, UPC has focused on providing high quality education oriented towards educating upstanding and innovative leaders with a global vision, so they can transform Peru.

Use Case Description

The application manages the student enrollment process at UPC. It facilitates the submission of enrollment applications for students interested in joining the university. Students can provide required information, such as their educational background, supporting documents, and contact details. Additionally, Students can pay their tuition and track the flow of their acceptance process.

All the data collected is presented through summaries of the registration process in dashboards to the company's senior managers.

Problem Statement

Due to the high demand during registration periods, the current system has suffered failures, causing many of the recurring students to be unable to enroll and running out of places. Therefore, the university presents failures in the continuity of the students of lower cycles.

These system crashes have generated claims in the payment processes and failed registrations. In this way, the data in the dashboards has been unrealistic and not very accurate.

3 Whys

Why do anything?

First, to contribute to the Students Experience and the institutional reputation by showing its commitment with the student satisfaction. Second, Addressing issues helps prevent disruptions in the enrolment process. Also, an improved system provides more reliable, real-time data that can be used for informed desitions.

3 Whys

Why Now?

Addressing the issues now ensures that the system is ready and running smoothly for the upcoming enrollment and registration cycles. Also, it will inmediatly enhance the experience and helps identify and address potential long-term problems. Early issue resolution often involves more efficient resource utilization. It prevent the lost of time and resources.

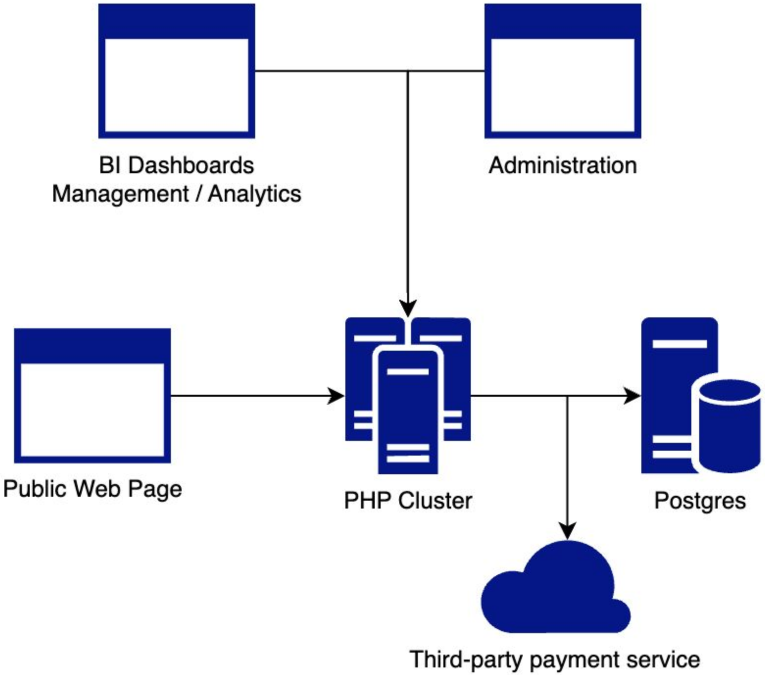
3 Whys

Why MongoDB?

MongoDB provides the infraestructure to centralice all the involved data in Atlas, It also reduces the architectural complexity of a high availability software.

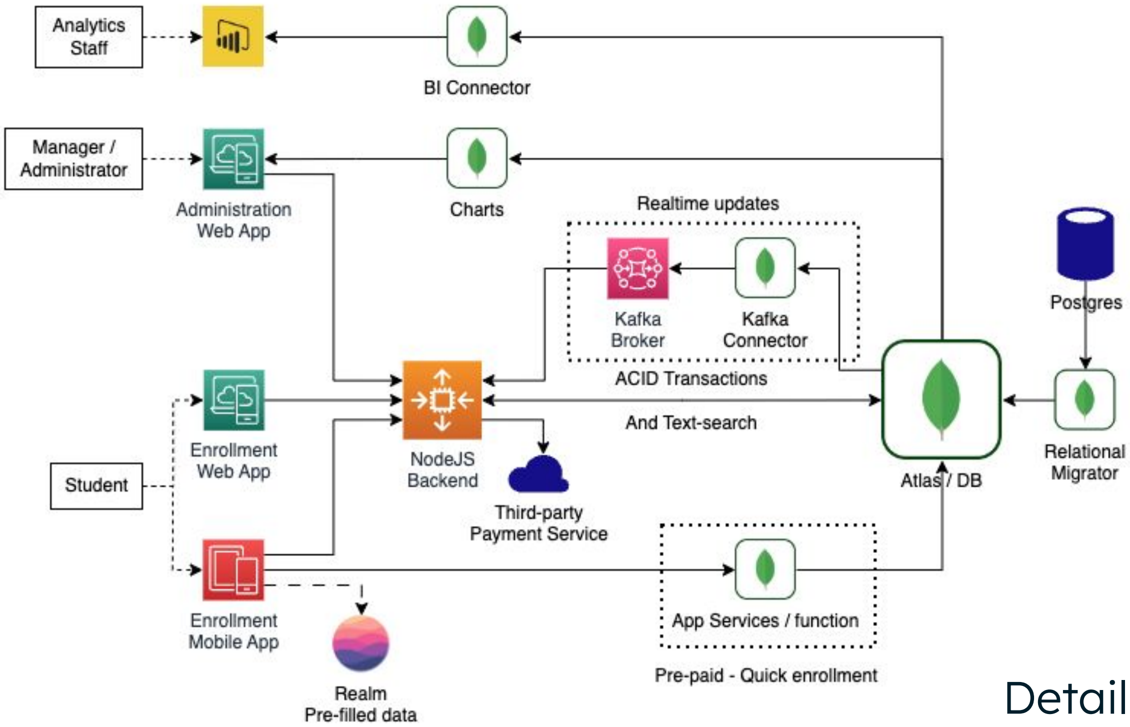
The multi cloud availability makes easier the management of the deployment and the database. MongoDB’s elasticity allows the client to not worry about resource allocation limits in case of a drastic increase in clients.

Current Architecture



Currently there's only one cluster on-premise that provides web applications using a LAPP Stack. And uses a third-party payment service through a REST API

Proposed Architecture



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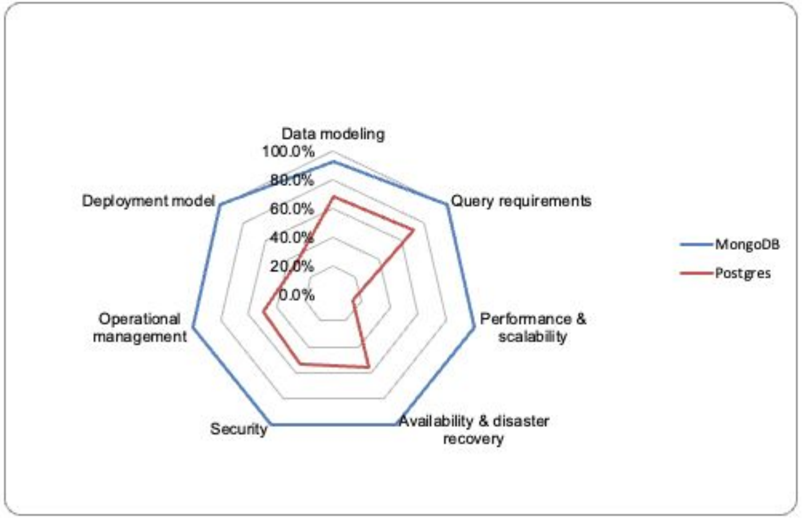
Proposed Architecture

The new architecture propose independent apps for management, enrollment and analytics. Administration app implements Charts and connects to the backend. Analytics Staff could query data in exploration data tools such as PowerBI. Enrollment app uses ACID Transactions in the enrollment process and text-search to find the available carreers and its details, the rest of the year this app works as a catalog of careers of the university.

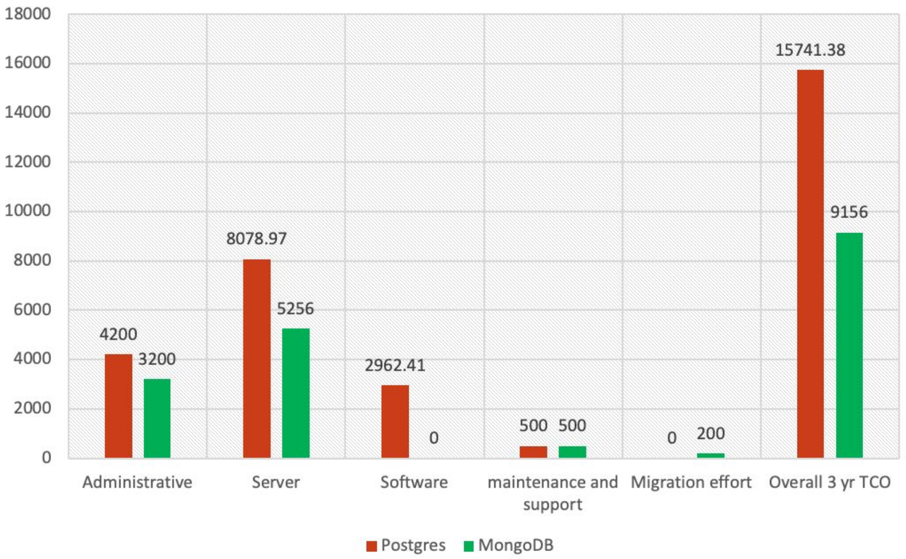
Additionally, enrollment is available through the mobile app. Provides a way to pre-populate enrollment by storing the information locally in Realm. Also, the student can quickly enroll (if previously paid) using App Services (Functions).

Modernization Scorecard

UPC - MongoDB Modernization Scorecard Results					
Category	**Maximum**	**MongoDB**	**Postgres**	**MongoDB**	**Postgres**
Data modeling	126	117	86	92.9%	68.3%
Query requirements	63	63	45	100.0%	71.4%
Performance & scalability	42	42	6	100.0%	14.3%
Availability & disaster recovery	27	27	15	100.0%	55.6%
Security	30	30	16	100.0%	53.3%
Operational management	30	30	15	100.0%	50.0%
Deployment model	6	6	2	100.0%	33.3%
Total	**324**	**315**	**185**	**97.2%**	**57.1%**



TCO/Pricing



Customer can save more than $5000 working with MongoDB.

Purchasing a server always increases initial expenses and does not allow for flexible scaling.

Instead, MongoDB allows customer to scale at any time and maintain a stable price from day one

Competition Comparison

Why not just migrate to the cloud? Well, Migrate would solve most of the problems of the current architecture. MongoDB and Postgres are both great Databases but in this situation MongoDB go the extra mile, and not just provides the Database. MongoDB provides a set of tools that not only solve the problem but also improve the capabilities of the current system.